

# IFQ557: Assignment 2 Submission Report

## FuturePath - Nonprofit Education Event Management System

### Section 1: Team declaration

#### Option A - Functional Team

| Functional Team               |                                                                                                                                 |                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Group/team number             | Group 07                                                                                                                        |                           |
| GitHub or Gitee name and link | <a href="https://github.com/7000frank/futurepath-event-management">https://github.com/7000frank/futurepath-event-management</a> |                           |
| Full name                     | Student #                                                                                                                       | Email                     |
| Frank                         | 2422441010                                                                                                                      | 2268109713@qq.com         |
| Chen                          | 2422441008                                                                                                                      | lv13913119139@outlook.com |
| George                        | 2422441020                                                                                                                      | yonghaopu67@gmail.com     |

#### Application overview

FuturePath upgrades the selected Assignment 1 prototype into a working Flask and SQLite application for free Gaokao guidance events. It provides dynamic event discovery, score-based recommendations, event details, comments, authentication, role-based event management, one-place bookings, booking history and automatic event states.

*Technology: HTML, CSS, Bootstrap, Python, Flask, Flask-WTF, WTForms, Flask-SQLAlchemy, SQLAlchemy, Flask-Login, Bootstrap-Flask, Werkzeug and SQLite.*

## Section 2: Implementation summary

| Assessment area                   | Implemented evidence                                                                                                                                              | Verification     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Landing page and search</b>    | Database-backed event cards; keyword, category and status filters; score finder; live state and remaining-place displays.                                         | Browser QA       |
| <b>Event details and comments</b> | Public event detail pages show image, description, date, time, venue and presenter. Logged-in users can post dated comments.                                      | Automated tests  |
| <b>Event management</b>           | Teacher accounts create events. Only the event owner can update or cancel. Open, Sold Out, Inactive and Cancelled states are calculated automatically.            | Permission tests |
| <b>Bookings</b>                   | Authenticated users reserve one free place, receive a unique reference, view history, cancel and rebook. Capacity and Sold Out state update from active bookings. | Booking tests    |
| <b>User management</b>            | Registration, Student/Teacher roles, password hashing, login, logout and route-level access control are implemented with Flask-Login and Werkzeug.                | Auth tests       |
| <b>Errors and validation</b>      | Custom 404/403/500 pages, CSRF-protected forms, WTForms validation, capacity checks, ownership checks and friendly flash messages.                                | Error tests      |

### Database design and supplied data

The SQLite database contains User, Event, Booking, Comment and Category tables with foreign-key relationships. The submitted database includes synthetic teacher and student accounts, event categories, Open/Sold Out/Inactive/Cancelled examples, comments and booking history. Prices are stored as 0.00 because FuturePath is nonprofit.

### Testing status

Twelve automated unittest cases pass. They cover registration, login/logout, role permissions, search, comments, creation and ownership, booking uniqueness, cancellation and rebooking, Sold Out behaviour, and custom 404/500 handling. Browser checks were also completed for desktop and mobile layouts without horizontal overflow or console errors.

## Section 3: Task contribution table

| Task/workstream                       | Contributor(s)                 | Completed work                                                                                                         |
|---------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Prototype and visual direction</b> | Frank                          | Selected and adapted the FuturePath Assignment 1 prototype; maintained the nonprofit theme and content direction.      |
| <b>Flask application structure</b>    | Frank                          | Application factory, blueprints, routes, templates, forms, static assets and package configuration.                    |
| <b>Database and seed data</b>         | Frank / Chen                   | SQLAlchemy models, relationships, calculated capacity/status properties and synthetic SQLite data.                     |
| <b>Authentication and roles</b>       | Frank / Chen                   | Registration, password hashing, login/logout, Student/Teacher roles and access-control checks.                         |
| <b>Discovery and event pages</b>      | Frank                          | Search, category/status filters, score finder, event details, comments and responsive presentation.                    |
| <b>Event management</b>               | Frank / Chen                   | Teacher creation workflow, ownership-based update/cancellation, validation and image upload.                           |
| <b>Booking workflow</b>               | Frank                          | One-place booking, unique references, duplicate prevention, history, cancellation, rebooking and Sold Out calculation. |
| <b>Testing and error handling</b>     | Frank / George                 | Automated tests, custom errors, browser QA, mobile checks and edge-case verification.                                  |
| <b>Report and submission package</b>  | Frank; Chen / George reviewing | Declaration, contribution evidence, AI statement, final review and ZIP assembly.                                       |

## Section 4: Individual contribution statements

*The following statements summarise each team member's contribution to the completed application.*

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frank</b>  | Frank led the FuturePath project from prototype selection through final delivery. He designed and refined the interface, built the core Flask application structure, integrated database-backed event discovery and score recommendations, and implemented the booking, cancellation, rebooking and event-status workflows. He coordinated authentication, ownership controls, responsive layouts, custom branding and event imagery, then carried out end-to-end debugging and browser verification. Frank also organised the final report, repository, test evidence and submission package, ensuring the solution remained consistent with the nonprofit concept and Assignment 2 requirements. |
| <b>Chen</b>   | Chen supported backend development and data persistence. He assisted with SQLAlchemy models and relationships, registration and role-based access control, event creation and ownership checks, comment storage, and the SQLite demonstration data. He reviewed WTForms validation, capacity calculations and status updates, and helped verify that database changes were reflected consistently across event pages and booking history. Chen also reviewed the completed Flask routes and contributed to functional checks of authentication and event-management workflows before final integration.                                                                                            |
| <b>George</b> | George contributed testing, requirements review and project coordination support. He checked the registration, login, search, comments, event-management and booking workflows against the task sheet, and helped verify custom error pages and edge cases. He tested responsive layouts at desktop and mobile widths, reviewed the supplied SQLite demonstration data, and recorded issues for correction. George also checked the repository and final ZIP structure, reviewed the contribution and AI declarations, and supported the final submission-quality review.                                                                                                                          |

## Section 5: git shortlog output

```
1    Frank <2268109713@qq.com>
```

## Section 6: Team use of AI statement

### Part B - AI was used in an authorised way

As a team, we report that AI tools were used in an authorised way. AI supported interpretation of the brief and rubric, drafting and refinement of Flask code, debugging of booking and permission workflows, interface and responsive-design review, test creation, and preparation of documentation. Team members remained responsible for reviewing the generated output, checking it against the supplied requirements, running the application and tests, and deciding which changes to retain.

### Representative AI use example 1

|                      |                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prompt</b>        | Use the FuturePath Assignment 1 prototype as the base and upgrade it into a working Flask + SQLite dynamic website. Include home/search, event details and comments, bookings and history, event management, registration/login, error handling and test data. |
| <b>Input/context</b> | Assignment 1 prototype ZIP, Assignment 2 brief and rubric, declaration template, existing HTML/CSS/images and the required technology stack.                                                                                                                   |
| <b>AI output</b>     | A draft Flask package with an application factory, models, forms, authentication, routes, templates, static assets, seed data and automated tests. The output was reviewed and iteratively corrected against the brief.                                        |
| <b>Team review</b>   | The team ran the website locally, checked database-backed workflows, revised roles and permissions, adjusted content and images, and retained only changes that matched the intended FuturePath design.                                                        |

### Representative AI use example 2

|                      |                                                                                                                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prompt</b>        | Each account should hold only one place per event. Cancellation must release the place, and a cancelled user should be able to book again as if booking for the first time. Booking success should remain on the event details page. |
| <b>Input/context</b> | Screenshots of the booking panel and history, the current Flask routes/models/templates, and the requested one-person nonprofit booking rule.                                                                                        |
| <b>AI output</b>     | Updates to duplicate-booking checks, cancellation/rebooking logic, booking IDs, capacity calculation, detail-page states and regression tests.                                                                                       |
| <b>Team review</b>   | The workflow was verified through automated tests and browser checks. The final interface shows Book My Place after cancellation, Already Booked after confirmation, and no forced redirect to My Bookings.                          |